# SIMATS ENGINEERING

## ASSESSMENTTOOL2:IndustryProblemSolving Task

**COURSE NAME:** Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1501

**COURSE OUTCOME COVERED**

**CO4:** *Analyzebigdatacharacteristicsand challenges across domains including security and application aspects.* (BL4)

Dave's Taxonomy: Articulation (Level 4)
Question Count: 5
Duration :60 Minutes

**Total Marks: 40**

**Code of Conduct**

I Nitheesh S192524329 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.
Signature of the student: Nitheesh

**Assessment Tasks Industry Problem Solving Task**

**Retail Data Optimization Problem -** A retail operates both physical stores and an online platform. It generates largevolumes of sales,inventory, and customer behavior data. The company wants real-time inventory tracking and demand forecasting. It also aims to deliver personalized marketing recommendations to customers. Design a big data architecture to support these requirements. Explain the tools, technologies, and analytics models you would use.

**Social Media Fraud Detection Problem -** A social media platform processes billions of userinteractions daily. Theseinclude posts,comments, likes, and shares. The platform wants to detectfake accounts and misinformation quickly. Real-time or near real-time processing is required. Designascalable big data solution for this use case. Explain the role of machine learning and stream processingframeworks.

**Government Data Platform Problem** A government agency is building a national data platform. Itcombines census, economic, and geospatial datasets. Different departments need access to shared and integrated data. The system must ensure interoperability across multiple agencies. Strict privacy and security requirements must be followed. Propose a governance and data management strategy.

**Banking Fraud Detection Problem**

Abank processes millions of financial transactions every minute. It needs to detect fraudulent activities in real time. The system must analyze historical and streaming transaction data. Low latency and high accuracy are critical requirements. Design a big data pipeline for fraud detection. Explain the technologies and algorithms you would implement.

**Agriculture Smart Farming Problem**

Asmartfarming system collects soil, weather, and crop data. Sensors and drones provide real-time field information. Farmers want insights to improve yield and resource usage. Data is generated from multiple distributed sources. Design a big data solution for precision agriculture. Explain how analytics can support decision-making.

**Industry Problem Solving Rubric**

| **Criteria** | **Weightage** | **Level 4 - Exemplary** | **Level 3 - Proficient** | **Level 2 - Developing** | **Level 1 - Beginning** |
|---|---|---|---|---|---|
| Understanding of Industry Context | 20% | Demonstrates strong understanding of real-world problem context | Good understanding | Basic understanding | Weak understanding |
| Application of Big Data Concepts | 25% | Applies highly relevant big data ideas and tools | Mostly appropriate application | Partial application | Weak application |
| Solution Design | 25% | Solution is practical, scalable, and well-reasoned | Reasonable solution | Basic solution with gaps | Unclear solution |
| Security / Risk Awareness | 15% | Strong awareness of security and operational risks | Reasonable awareness | Limited awareness | Poor awareness |
| Clarity and Professionalism | 15% | Well-structured and professional response | Generally clear | Moderately clear | Poorly presented |

**Total Marks Awarded:**

## Q1-Retail Data Optimization

Aretail companygenerates large volumes of **sales, inventory, and customer behavior data** from physical stores and its online platform. A scalable Big Data architecture can integrate these different data sources and process them in real time.

The proposed architecture collects data from **POS systems, e-commerce transactions, inventory systems, customer interactions, and website/mobile applications**. The data can be stored in a scalable cloud data lake or distributed storage system. **Apache Kafka** can be used for real-time data streaming, while **Apache Spark/PySpark** can process both real-time and historical data.

For **real-time inventory tracking**, streaming data from stores and online orders is processed continuously to update stock levels. This helps the company identify low-stock products and avoid overstocking or stock-outs.

For **demand forecasting**, historical sales data can be analyzed using machine learning models such as **Linear Regression, Random Forest, or time-series forecasting** to predict future product demand. The predicted demand can help the company plan inventory and restocking.

For **personalized marketing**, customer behavior such as purchases, searches, and product interactions can be analyzed using **recommendation and customer-segmentation techniques**. For example, a recommendation model can suggest products based on a customer's previous purchases and browsing behavior.

## Architecture

POS / Online Store / Inventory / Customer Data
↓
Apache Kafka
↓
Real-Time Data Stream
↓
Apache Spark
↓
Cloud Data Lake / Storage
↓

↓ ↓ ↓

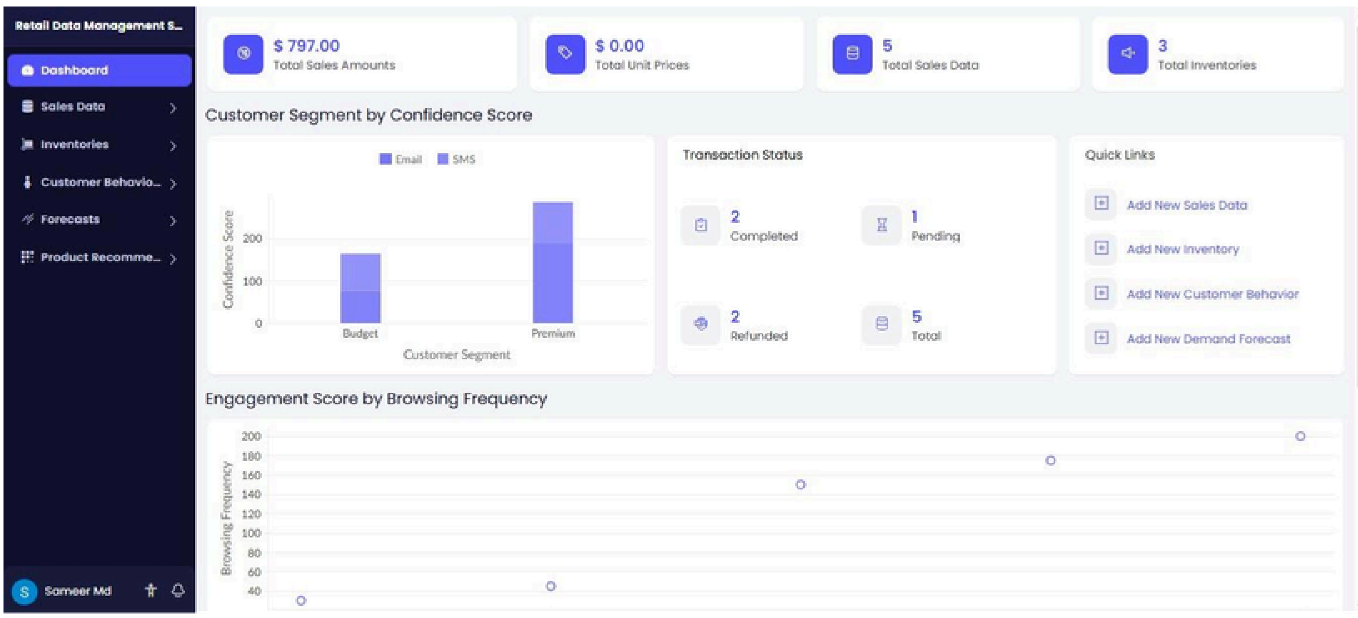
Inventory Tracking | Demand Forecasting | Customer Analytics

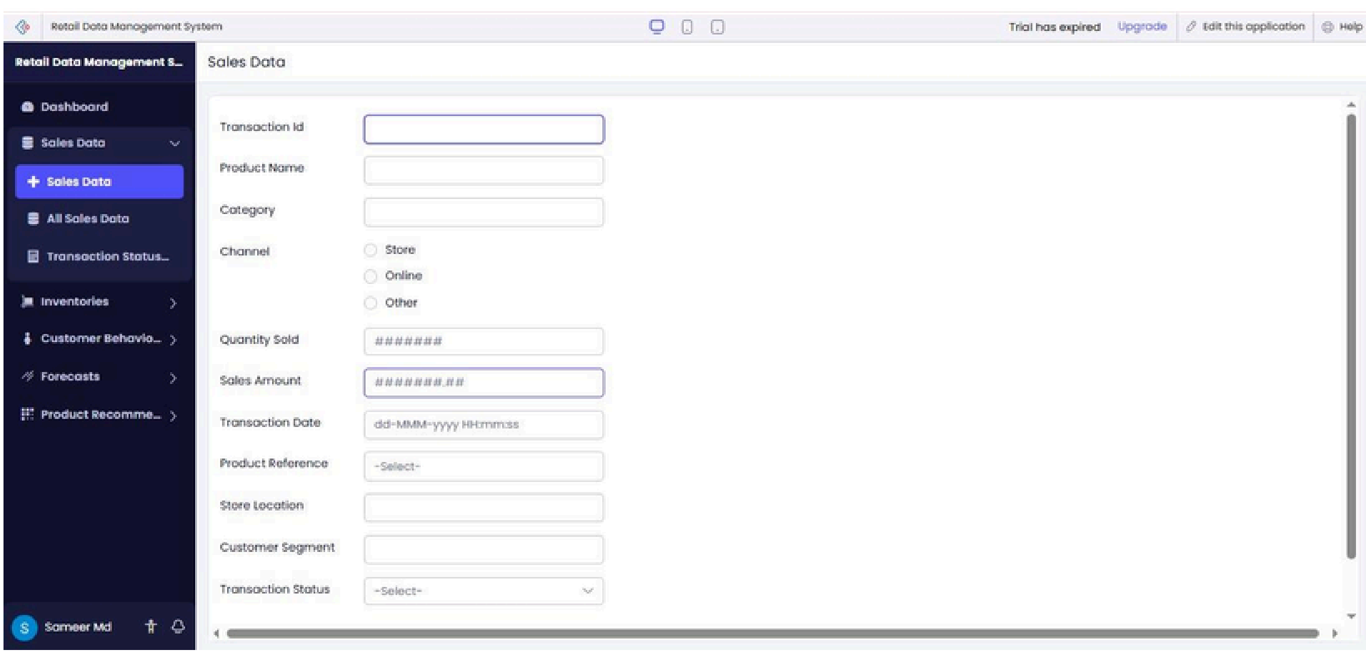
↓ ↓ ↓

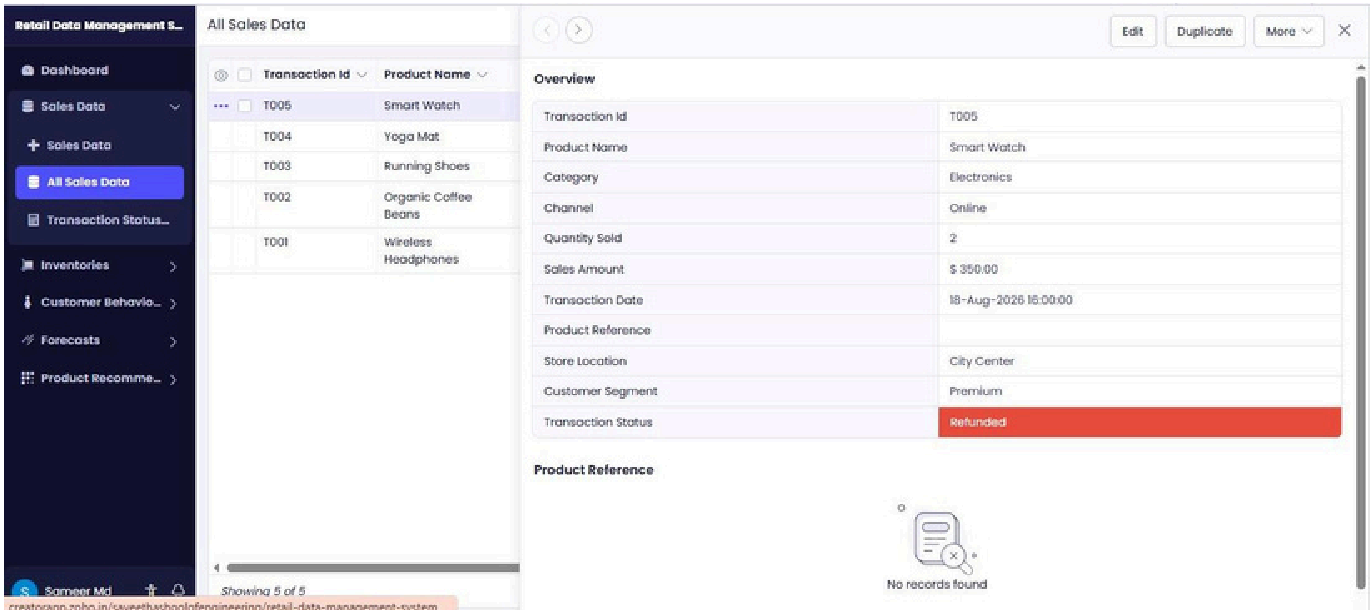
↓
Business Dashboard
↓
Better Retail Decisions

This architecture is **scalable** because distributed processing can handle increasing volumes of retail data. It also supports real-time decision-making and can be expanded as the number of stores, customers, and transactions increases.

For security, the company should implement **authentication, role-based access control, encryption, secure data transmission, and monitoring** to protect customer and business data. Overall, the Big Data solution helps the retailer maintain accurate inventory, forecast demand, understand customer behavior, and provide personalized recommendations. This improves operational efficiency, reduces stock-related problems, and supports better customer service. The solution directly addresses the assessment requirements for real-time inventory tracking, demand forecasting, personalized marketing, and Big Data architecture.






## Q2 – Social Media Fraud Detection

AscalableBig Datafrauddetection system can collect data from **posts, comments, likes, shares, user profiles, login activities, and interaction patterns**. Since the data is generated continuously, a stream-processing architecture can be used for near-real-time analysis. **Apache Kafka** can collect and stream user interactions from different sources. **Apache Spark Streaming/PySpark** can process these streams and extract useful features such as posting frequency, number of interactions, repeated content, unusual login patterns, and account activity. Machine learning can then be used to identify suspicious accounts and misinformation. **Classification models such as Random Forest or Logistic Regression** can classify accounts as genuine or suspicious based on their behavior. For misinformation detection, text-analysis techniques can examine posts and identify suspicious or misleading content.

The system can store historical data in distributed cloud storage and use it to train and improve machine learning models. Real-time processing can compare new user activity with previously identified patterns and generate alerts when suspicious behavior is detected.

For security, the platform should use **authentication, access control, encryption, and monitoring** to protect user data. The solution should also be scalable so that additional processing resources can be added when the number of social-media interactions increases.

## Architecture

Posts / Comments / Likes / Shares / User Activity

↓

Apache Kafka

↓

Stream Processing

↓

Apache Spark

↓

Feature Extraction

↓

Machine Learning Models

↓ ↓

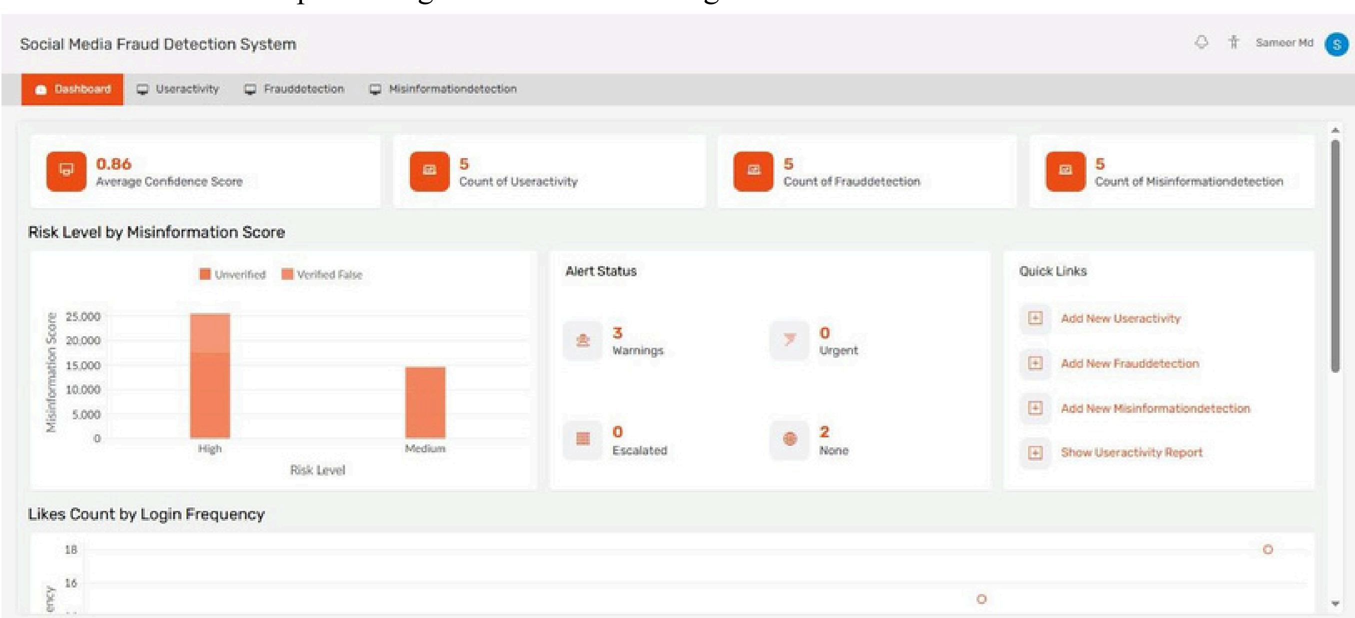
Fake Account Detection | Misinformation Detection

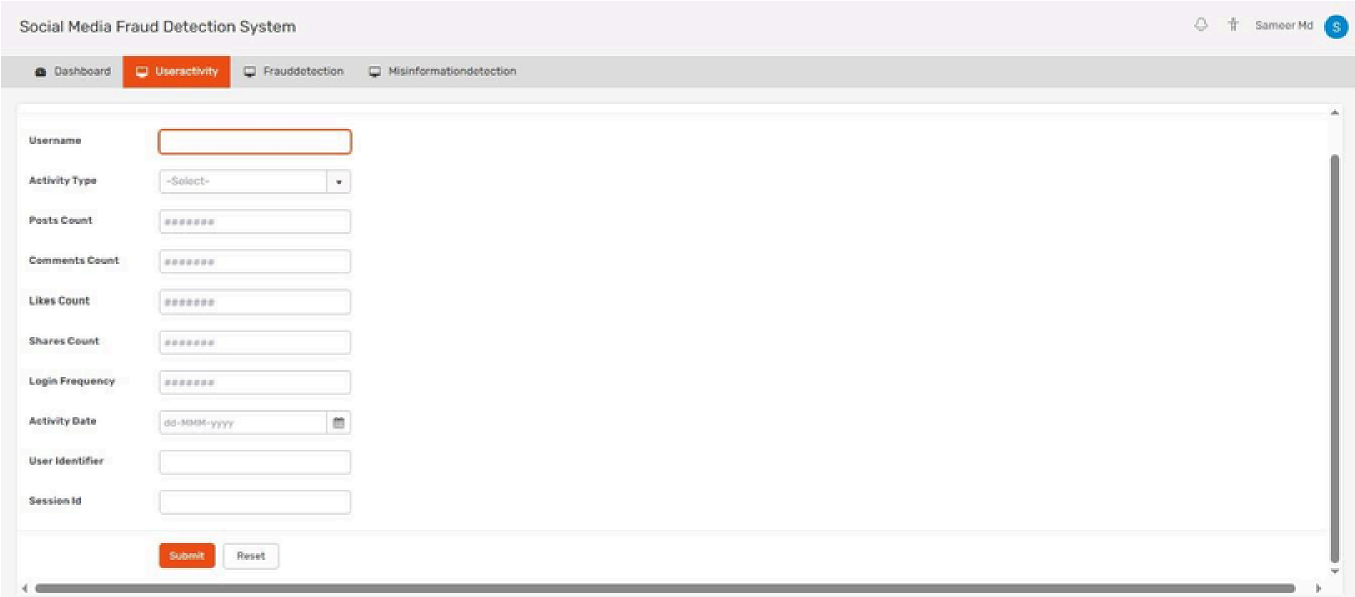
↓ ↓

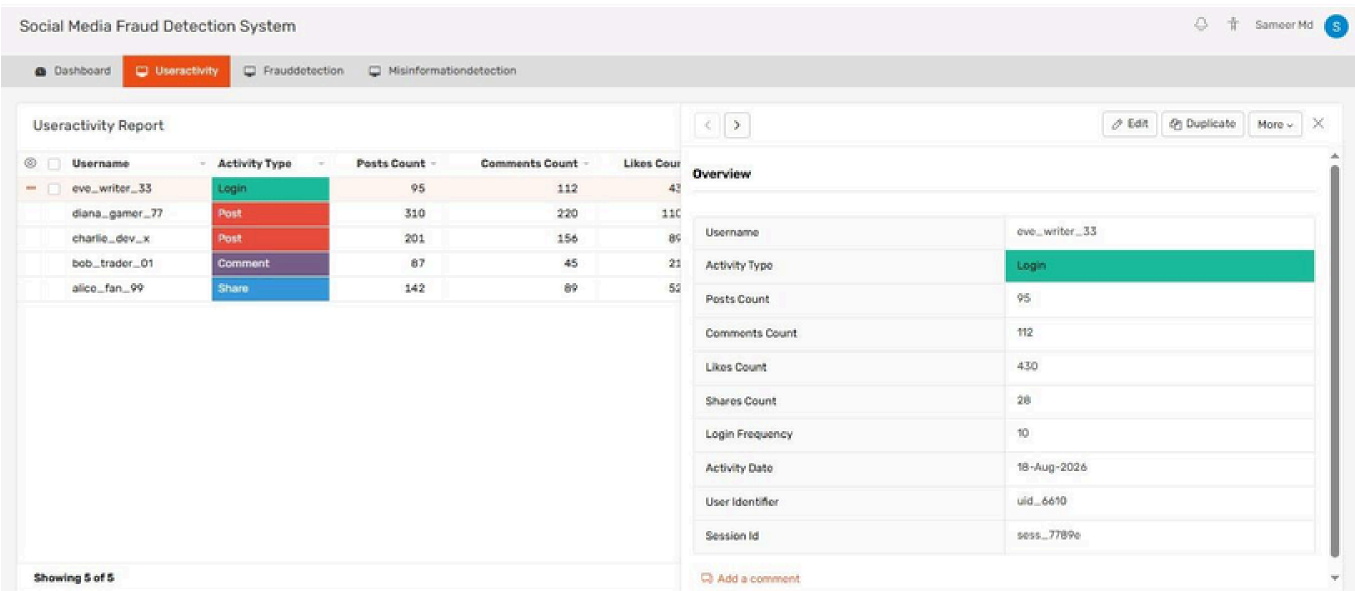
↓

Alert / Action

Thus, combining **Kafka for streaming, Spark for distributed processing, and machine learning for detection** provides a scalable solution for detecting fake accounts and misinformation with low processing delay. This directly addresses the assessment requirement for real-time/near-real-time processing and machine-learning-based fraud detection.






# Q3 – Government Data Platform

A government agencyis developinga**national data platform** that combines large datasets such as **census data, economic data, and geospatial data**. Since different government departments need to access and share this information, the platform should provide a common and integrated data environment.

A **centralized Big Data platform** can be used to collect, store, integrate, and process data from multiple government agencies. Data can be stored in a scalable distributed storage system and processed using Big Data technologies. Standard data formats and APIs should be used to ensure **interoperability**, allowing different departments and systems to exchange and use data effectively.

A strong **data governance strategy** should define rules for data ownership, quality, access, privacy, security, and usage. Each department should have clearly defined access permissions based on its role. Sensitive citizen information should be protected using **encryption, authentication, role-based access control, and secure data transmission**.

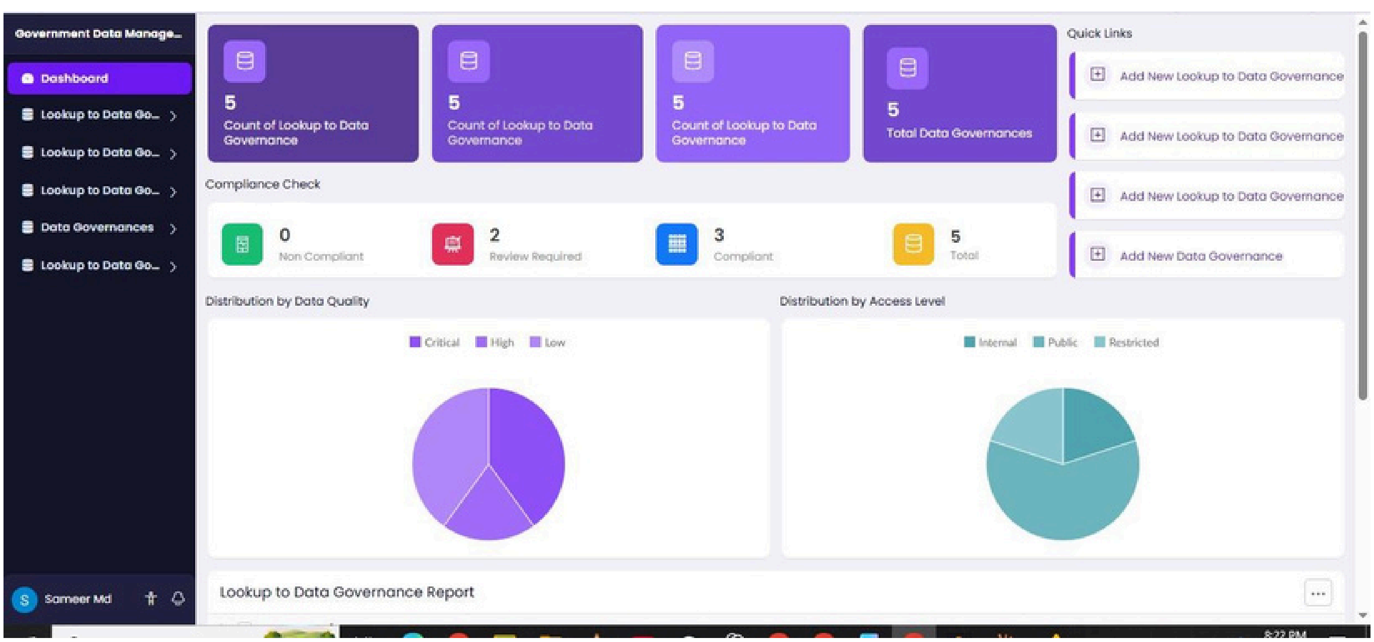
The platform should also maintain **data catalogs and metadata** so that departments can understand the source, format, meaning, and quality of datasets. Data quality checks should be performed before integrating information from different agencies.

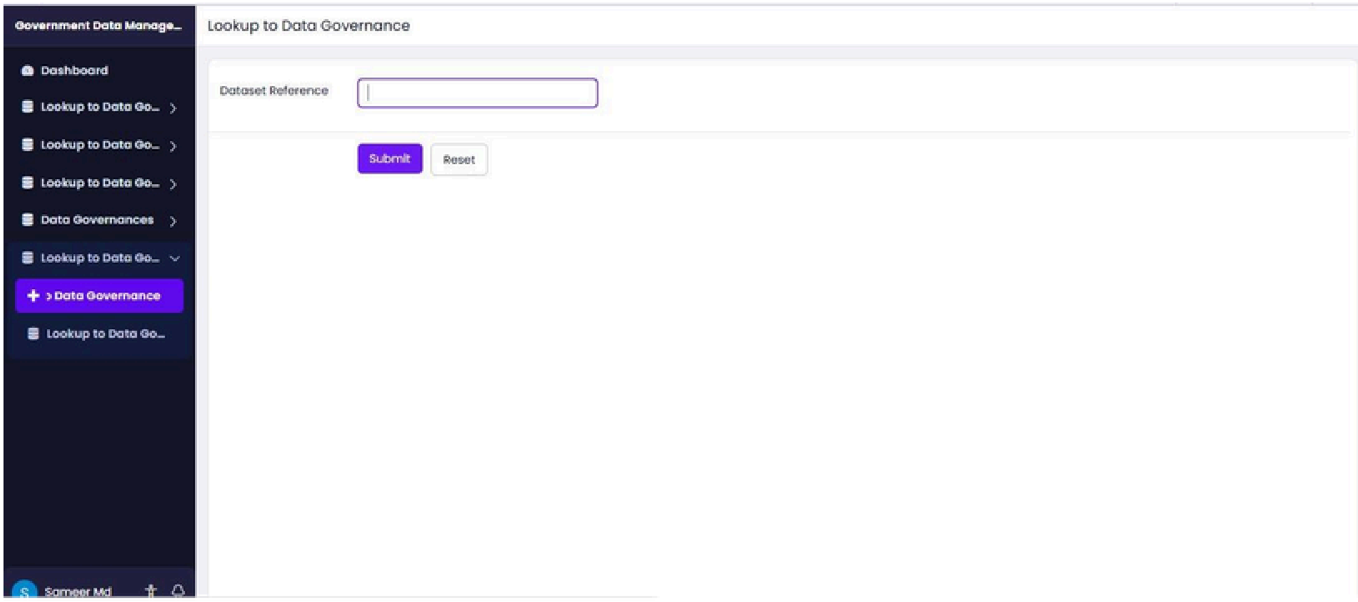
## Proposed Architecture

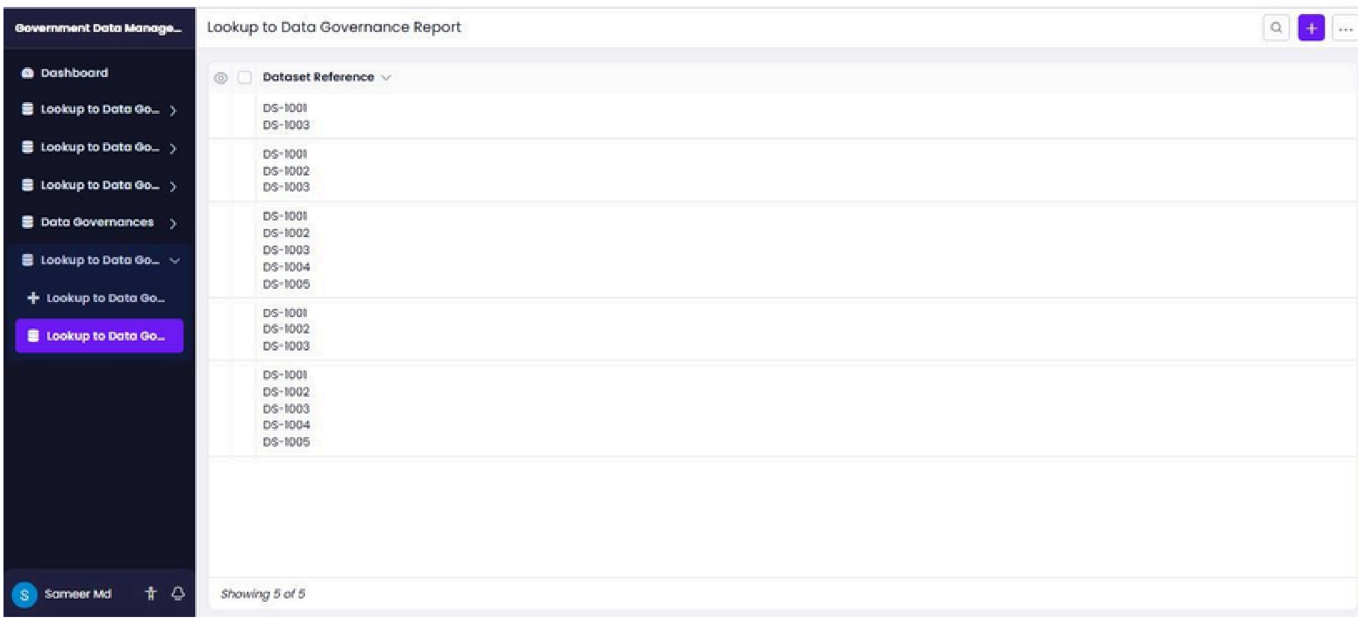
```
Census Data ──────┐
Economic Data ────┼──→ Data Integration
Geospatial Data ──┘          ↓
                    Central Data Platform
                             ↓
                    Distributed Data Storage
                             ↓
                       Data Processing
                             ↓
            ┌────────────────┼────────────────┐
                 ↓           ↓           ↓
          Department A  Department B  Department C
              ↓              ↓              ↓
          └────────── Secure Access ──────────┘
```

The platform should implement **logging and monitoring** to track data access and detect unauthorized activities. Regular audits can help ensure that departments follow security and privacy policies.

Therefore, a centralized and governed Big Data platform can provide **interoperability, secure data sharing, data quality, privacy, and controlled access** across government agencies. This approach allows departments to use integrated census, economic, and geospatial information while maintaining strict security and privacy requirements. The solution directly addresses the assessment requirements for interoperability, privacy, security, governance, and integrated national data management.






## Q4 – Banking Fraud Detection

Abankcan implement a scalable **BigData fraud detection pipeline** to analyze both historical transaction data and real-time transactions. Transaction information such as account number, transaction amount, location, time, device information, and transaction type can be collected continuously.

**Apache Kafka** can be used to receive and stream transactions in real time. **Apache Spark Streaming/PySpark** can process these transactions with low latency and extract important features such as unusual transaction amounts, transaction frequency, location changes, and abnormal spending patterns.

Historical transaction data can be stored in distributed cloud storage and used to train machine learning models. Algorithms such as **Random Forest, Logistic Regression, or anomaly-detection techniques** can be used to classify transactions as genuine or potentially fraudulent. The trained model can then analyze incoming transactions and generate a fraud score.

If a transaction is identified as highly suspicious, the system can generate an immediate alert for further verification. Continuous monitoring and model updates can improve detection accuracy over time.

## Big Data Pipeline

Historical Transactions + Real-Time Transactions

↓

Apache Kafka

↓

Spark Streaming

↓

Feature Extraction

↓

Machine Learning Model

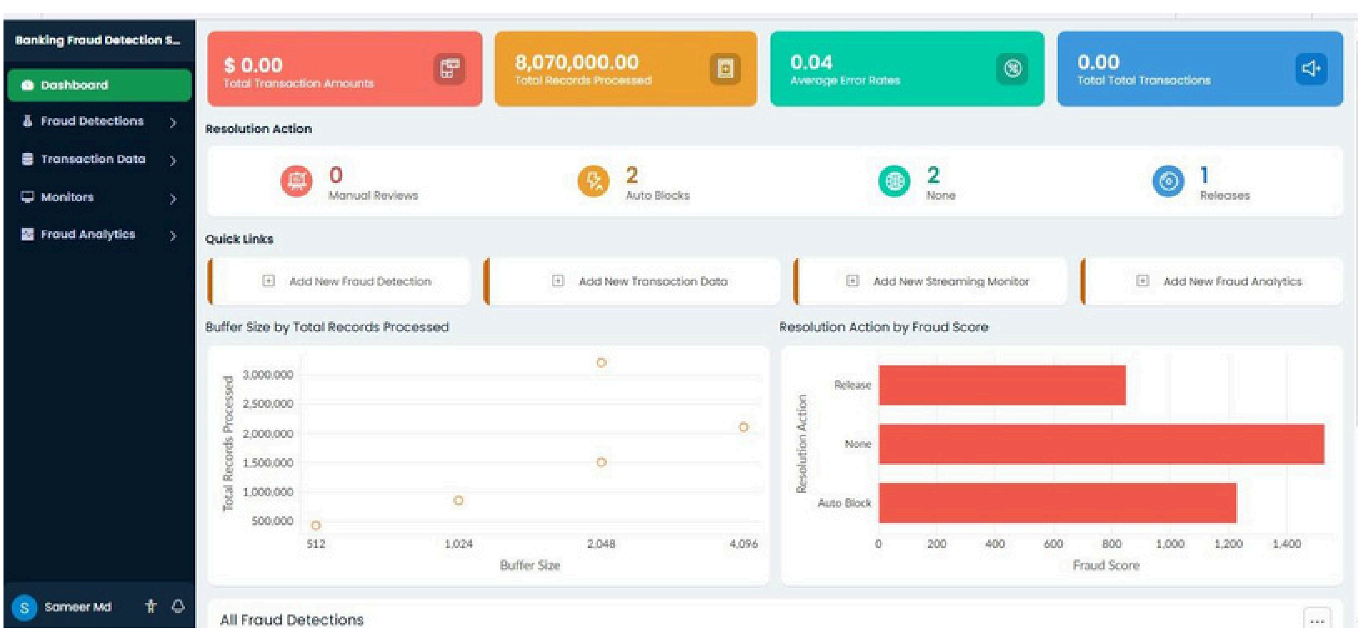
↓

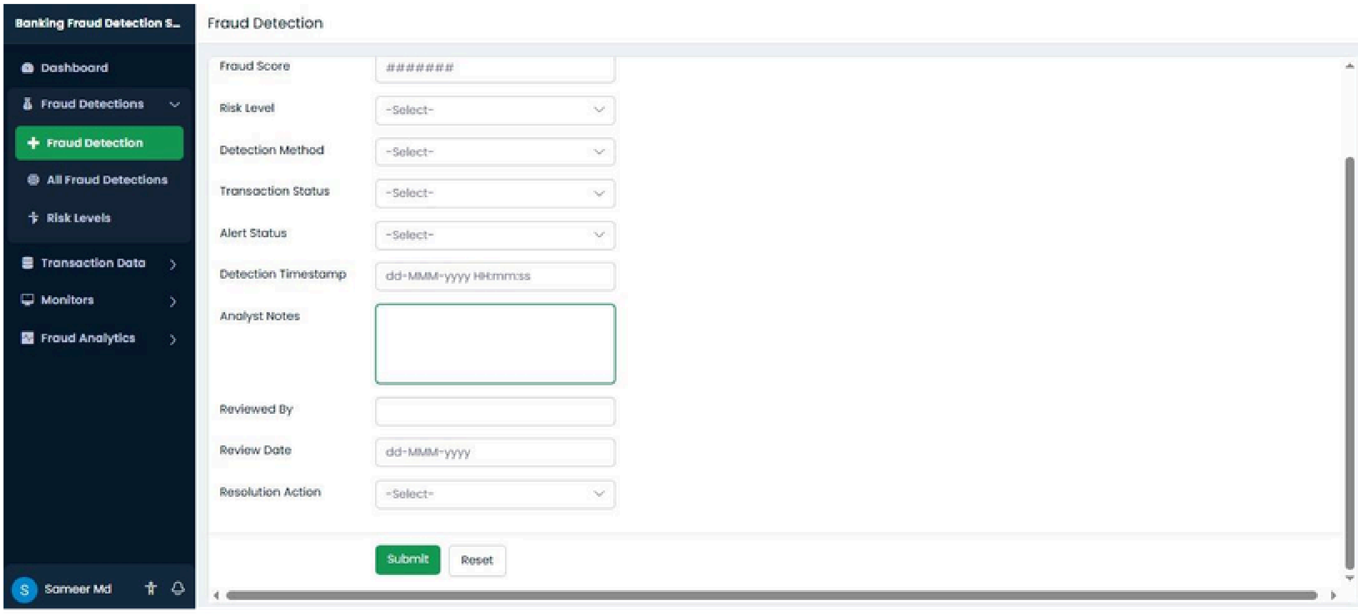
↓ Genuine Transaction ↓ Fraudulent

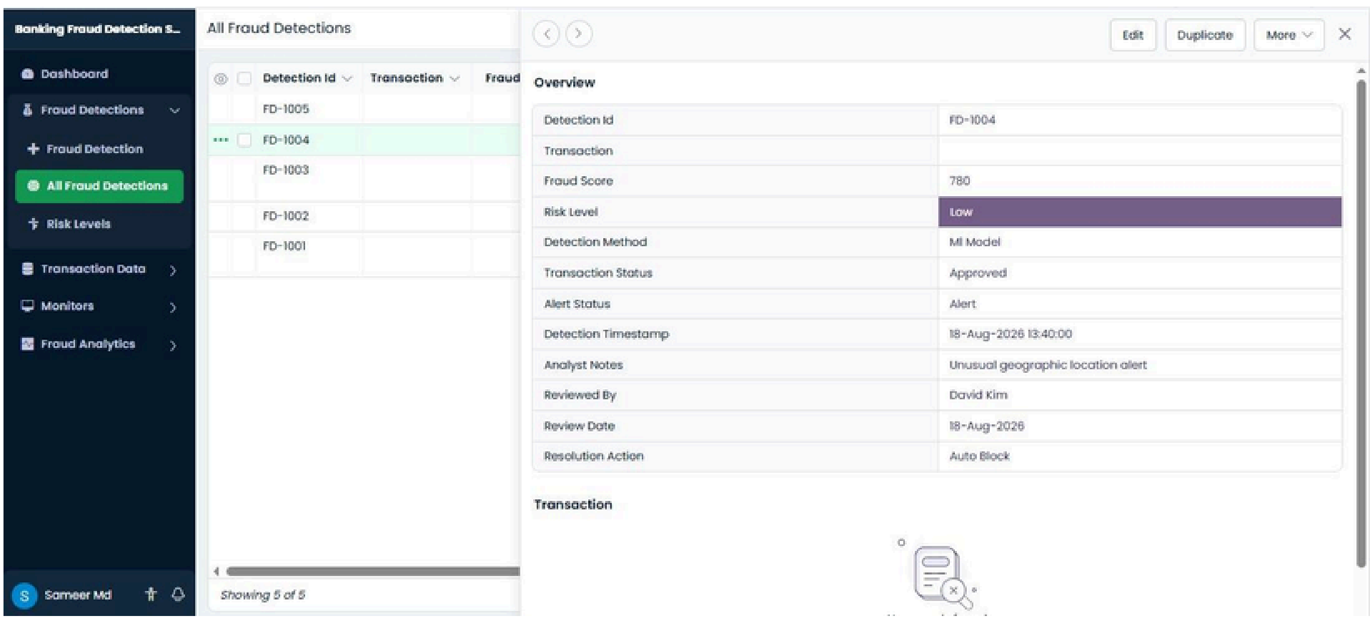
↓ Process Normally ↓ Alert / Block

The system should also use **encryption, authentication, access control, logging, and monitoring** to protect sensitive financial information. A distributed architecture allows the system to scale when transaction volume increases.

Therefore, combining **Kafka for real-time streaming, Spark for distributed processing, and machine learning for fraud detection** provides a scalable, low-latency solution for detecting fraudulent banking transactions while maintaining high accuracy. This directly addresses the assessment requirements for historical and streaming data, low latency, high accuracy, technologies, and algorithms






## Q5 – Agriculture Smart Farming

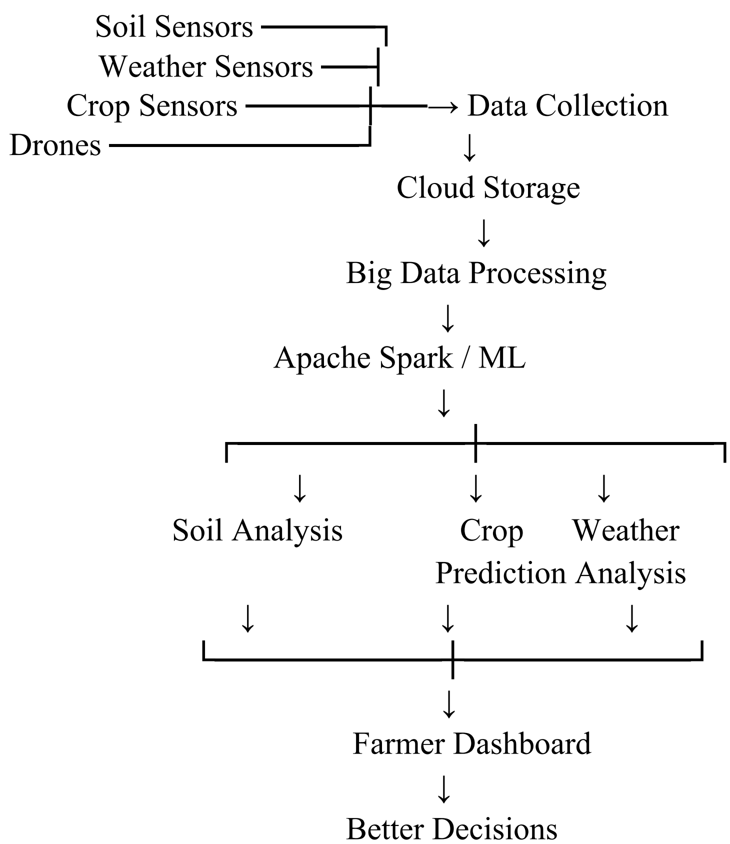
Asmart farming systemcan usea **BigData architecture** to collect and analyze data from soil sensors, weather stations, drones, and crop monitoring systems. Since the data comes from multiple distributed sources, it can be collected through IoT devices and stored in scalable cloud storage.

Real-time sensor data can be processed using **stream-processing technologies**, while historical agricultural data can be processed using distributed Big Data tools such as **Apache Spark**. The collected data can include soil moisture, temperature, humidity, rainfall, crop health, and nutrient levels.

Analytics can be used to identify patterns and provide useful insights to farmers. For example, **predictive analytics** can estimate crop yield and identify possible crop diseases. Soil and weather data can help determine the appropriate amount of irrigation and fertilizer required.

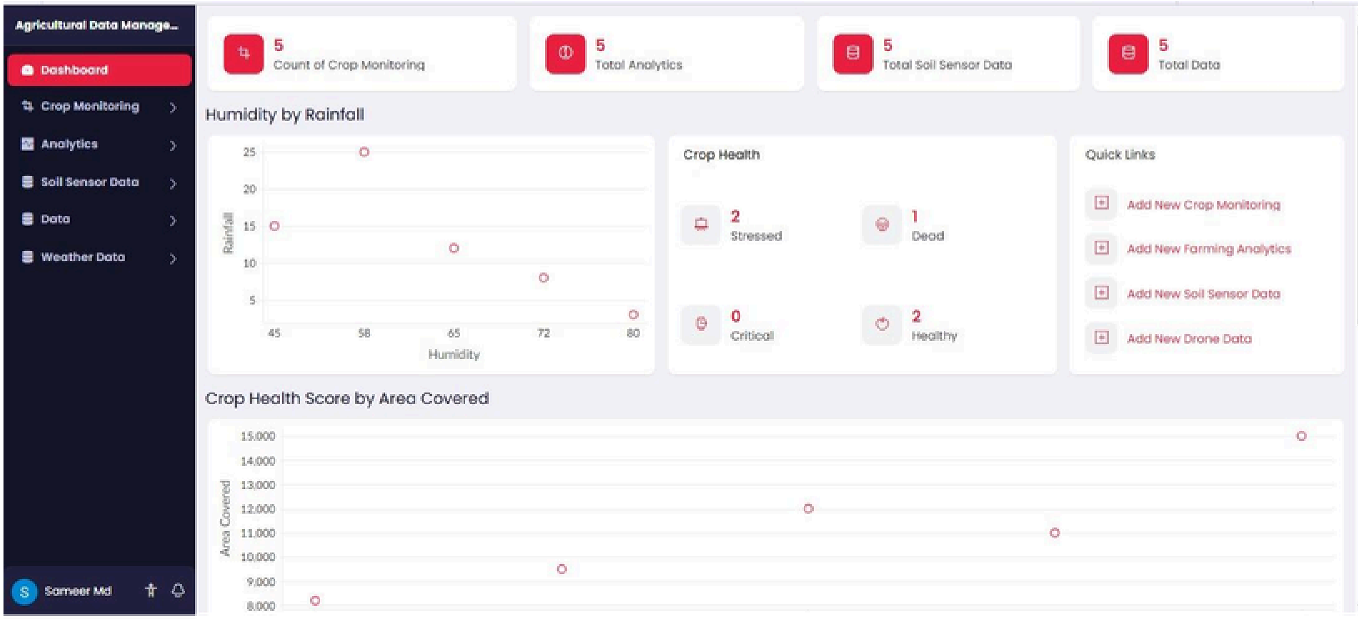
Drone images can also be analyzed to identify unhealthy crop areas. Based on these insights, farmers can make better decisions about **irrigation, fertilizer usage, pest control, and harvesting**. This can improve crop yield while reducing unnecessary use of water and other resources.

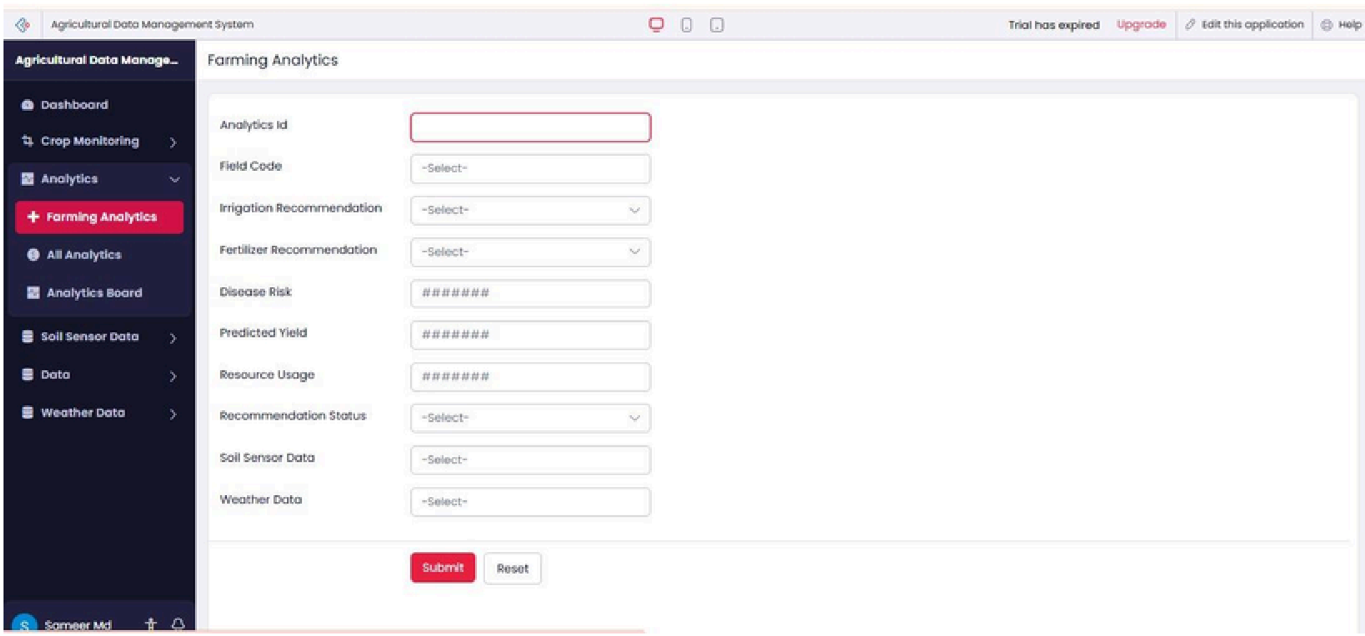
## Big Data Architecture

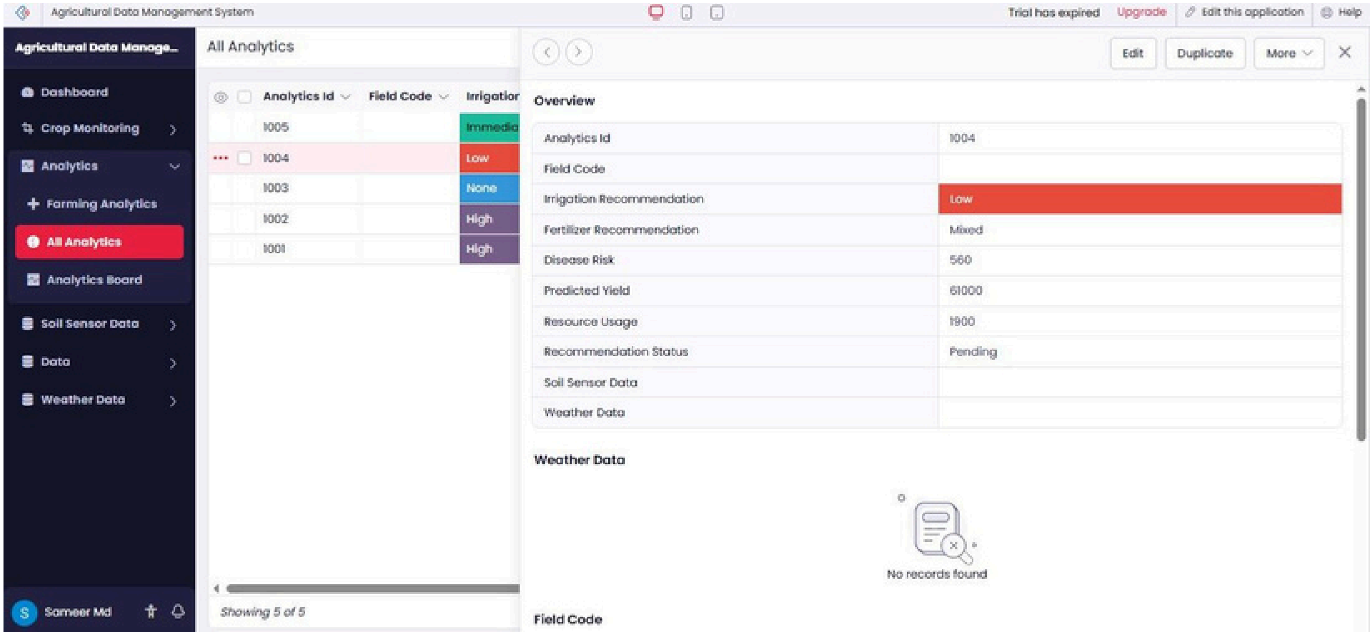



The system should also provide **secure data access, authentication, monitoring, and controlled sharing** to protect agricultural data.

Therefore, Big Data analytics enables precision agriculture by combining distributed sensor and drone data to provide **real-time insights, crop predictions, optimized irrigation, efficient resource usage, and improved decision-making**. This directly addresses the assessment requirement for distributed agricultural data and analytics-based decision support.


Agricultural Data Manage...
Dashboard
Crop Monitoring
Analytics
Soil Sensor Data
Data
Weather Data
Sameer Md
5
Count of Crop Monitoring
5
Total Analytics
5
Total Soil Sensor Data
5
Total Data
Humidity by Rainfall
Rainfall
Humidity
Crop Health
2
Stressed
1
Dead
0
Critical
2
Healthy
Quick Links
Add New Crop Monitoring
Add New Farming Analytics
Add New Soil Sensor Data
Add New Drone Data
Crop Health Score by Area Covered
Area Covered


Agricultural Data Management System
Trial has expired
Upgrade
Edit this application
Help
Agricultural Data Manage...
Dashboard
Crop Monitoring
Analytics
Farming Analytics
All Analytics
Analytics Board
Soil Sensor Data
Data
Weather Data
Sameer Md
Farming Analytics
Analytics Id
Field Code
Irrigation Recommendation
Fertilizer Recommendation
Disease Risk
Predicted Yield
Resource Usage
Recommendation Status
Soil Sensor Data
Weather Data
Submit
Reset


Agricultural Data Management System
Trial has expired
Upgrade
Edit this application
Help
Agricultural Data Manage...
Dashboard
Crop Monitoring
Analytics
Farming Analytics
All Analytics
Analytics Board
Soil Sensor Data
Data
Weather Data
Sameer Md
All Analytics
Analytics Id
Field Code
Irrigation
1005
Immedia
1004
Low
1003
None
1002
High
1001
High
Showing 5 of 5
Edit
Duplicate
More
Overview
Analytics Id
1004
Field Code
Irrigation Recommendation
Low
Fertilizer Recommendation
Mixed
Disease Risk
560
Predicted Yield
61000
Resource Usage
1900
Recommendation Status
Pending
Soil Sensor Data
Weather Data
Weather Data
No records found
Field Code